

SANTO HOTEL

PHASE 1

Product Understanding & Business Workflow

INTERN TRAINING & IMPLEMENTATION GUIDE

Project	Santo Hotel — Single-Tenant Hotel Booking Web Application
Phase	1 — Product Understanding
Audience	Software Engineering Intern / Junior Developer
Purpose	Understand the product and business workflow before coding
Suggested Duration	1–3 working days
Prerequisite	Read the Santo Hotel PRD and overall development workflow
Mobile App	Not part of this phase

Training principle: do not write production code until you can explain what the system is supposed to do.

1. Phase Overview

Phase 1 is the foundation of the Santo Hotel project. The purpose is not to build software yet. The purpose is to make sure the intern understands the business problem, users, booking lifecycle, major system components and rules the application must enforce.

Hotel booking systems are especially sensitive because availability, reservations, payments and cancellations interact with one another. A developer who does not understand the workflow can build a good-looking system that produces incorrect bookings.

PHASE 1 GOAL — By the end of this phase, the intern should be able to explain the Santo Hotel booking system from the perspective of a guest, hotel staff member and administrator without looking at the code.

2. Learning Objectives

- Explain what Santo Hotel's web application is designed to accomplish.
- Identify the main users of the system.
- Explain Guest, Staff and Administrator responsibilities.
- Describe the complete guest booking journey from search to confirmation.
- Describe the hotel staff workflow for managing a reservation.
- Explain what room availability means in a hotel booking system.
- Explain why a reservation is different from a physical room.
- Identify the major entities that will eventually become database structures.
- Explain the difference between room status, booking status and payment status.
- Identify business rules that must be enforced by the backend.
- Identify public and protected areas of the application.
- Translate a business requirement into a clear technical requirement.

3. Understand the Product

3.1 What Are We Building?

Santo Hotel needs a single-tenant hotel booking web application. Single-tenant means the application is built specifically for Santo Hotel rather than being a platform where multiple hotels create separate accounts.

PUBLIC / GUEST SIDE

Guests use the website to:

Search → View rooms → Book → Pay → Receive confirmation

PRIVATE / ADMIN SIDE

Hotel personnel use the administration system to:

Manage rooms → Manage availability → Manage reservations

→ Manage guests → Manage payments → View reports

3.2 What Problem Does It Solve?

- Guests should not need to call the hotel for every reservation.
- Hotel staff should have a central place to see reservations.
- The system should reduce accidental double bookings.
- The hotel should control room information and pricing.
- Guests should receive clear confirmation after booking.

- The hotel should have a reliable digital record of reservations and payments.

4. Understand the Users

4.1 Guest

The guest is the customer who wants to stay at Santo Hotel. They may browse without an account or use an account.

- Search available rooms.
- View room information.
- Select dates and number of guests.
- Enter personal information.
- Make payment.
- Receive booking confirmation.
- View or manage eligible bookings.

4.2 Hotel Staff

Staff operate the hotel reservation process. They may receive bookings from the website, phone calls, walk-ins or other channels.

- Search reservations.
- Create manual reservations.
- Modify reservations.
- Check guests in and out.
- Add operational notes.
- View room availability.

4.3 Administrator

The administrator manages configuration and operational data.

- Manage users and permissions.
- Manage rooms and room types.
- Manage rates and promotions.
- Manage reservations.
- View reports.
- Manage website content.
- Manage system settings.

5. Understand the Guest Booking Journey

5.1 High-Level Flow

```
Guest visits website
↓
Enters dates and guests
↓
System checks availability
↓
Available rooms are displayed
↓
```

```
Guest selects a room
↓
System calculates booking price
↓
Guest enters personal information
↓
Guest proceeds to payment
↓
Payment is verified
↓
Booking becomes confirmed
↓
Guest receives confirmation
```

5.2 Why the Order Matters

Availability should be checked again when the booking is being finalized. The browser may show a room as available, but another guest could have booked it before the first guest completes checkout.

TRAINING QUESTION — If two guests try to book the last room at the same time, what part of the system should prevent the room from being sold twice? The intern should identify the backend/database transaction and booking validation as the authoritative control.

6. Understand Hotel Room Availability

Availability is one of the most important concepts in the project.

6.1 A Room Is Not Always Available

Physical Room 101

```
Available
↓
Reserved
↓
Occupied
↓
Cleaning
↓
Available
```

A physical room can also be unavailable because it is under maintenance or out of service.

6.2 Date Availability

Existing reservation:
15 Oct → 18 Oct

Guest searches:
16 Oct → 17 Oct

Result:
Room 101 is NOT available

6.3 Back-to-Back Bookings

Booking A:
15 Oct → 18 Oct

Booking B:
18 Oct → 20 Oct

Potentially valid:

YES

Reason:

The first guest checks out on the 18th
and the next booking starts on the 18th.

The exact hotel check-in/check-out policy should be confirmed with Santo Hotel, but the intern must understand date-range overlap.

7. Understand the Three Important Status Groups

7.1 Room Status

Available
Reserved
Occupied
Cleaning
Maintenance
Out of Service

Room status describes the operational condition of a physical room.

7.2 Booking Status

Pending
Confirmed
Checked In
Checked Out
Cancelled
No Show

Booking status describes what is happening with the reservation.

7.3 Payment Status

Pending
Paid
Failed
Refunded
Partially Refunded (if supported)

Payment status describes the financial transaction. These statuses must not be confused with booking status.

TRAINING QUESTION — Can a booking be Confirmed while payment is Pending? This depends on the hotel's approved payment policy. The intern must identify it as a business rule, not guess in code.

8. Identify the Core Business Entities

Before database design begins, the intern should understand the objects that exist in the business.

Entity	Meaning
Hotel	Santo Hotel and its core information
Guest	Person making or associated with a reservation
User	Authenticated system user
Room Type	Category such as Deluxe King or Executive Suite
Room	Physical hotel room with a room number
Amenity	Feature such as Wi-Fi, TV or air conditioning

Entity	Meaning
Rate	Price/rate applicable to a room type and period
Booking	Reservation made by a guest or staff
Booking Room	Specific room allocation associated with a booking
Payment	Financial transaction related to a booking
Promotion	Discount or promotional rule
Notification	Message sent to guest or staff
Audit Log	Record of important administrative actions

9. Important Concept: Room Type vs Physical Room

Room Type

Deluxe King Room
Capacity: 2 adults
Bed: King
Base price: RM 450/night

Physical Room

Room 101 → Deluxe King
Room 102 → Deluxe King
Room 103 → Deluxe King

One room type can therefore have many physical rooms.

Deluxe King Room Type

■
■■■■■■■■■■

↓ ↓ ↓

Room 101 Room 102 Room 103

TRAINING QUESTION — If the hotel has 10 Deluxe King rooms, should the system create 10 different room types? No. Normally there is one room type with 10 physical rooms.

10. Convert Business Requirements into Technical Requirements

The intern should learn to translate a statement from a hotel manager into a technical requirement.

Example 1

Business:

Guests must not be able to book a room that is already reserved.

Technical:

The availability service must detect overlapping reservations before a booking can be confirmed.

Example 2

Business:

Guests can cancel up to 48 hours before arrival.

Technical:

The backend needs a cancellation policy and must calculate

whether a cancellation request falls inside the permitted window.

Example 3

Business:

Staff should know which rooms are under maintenance.

Technical:

Physical rooms need a status field and the availability engine must exclude maintenance rooms from sellable inventory.

Intern Exercise

Write the technical interpretation for these statements:

- A guest can book two rooms for the same dates.
- A Deluxe King can accommodate a maximum of two adults.
- Staff can create a booking for a guest over the phone.
- The hotel wants to offer a 10% promotional discount.
- A payment failure must not result in a falsely paid booking.

11. Identify Rules That Belong in the Backend

Anything affecting money, availability, authorization or data integrity should be enforced by the backend.

- Room availability.
- Maximum occupancy.
- Booking date validation.
- Price calculation.
- Discount calculation.
- Cancellation rules.
- Booking status changes.
- Payment verification.
- User permissions.
- Duplicate booking prevention.

RULE — The frontend can improve the user experience, but the backend is the final authority for business rules.

12. Public vs Protected System Areas

Public

- Home page
- Room listing
- Room details
- Availability search
- Hotel information
- Contact information
- Booking initiation

Protected Guest Area

- Profile
- Booking history
- Upcoming reservations
- Account settings

Protected Staff/Admin Area

- Admin dashboard
- Reservations
- Room management
- Rates
- Guests
- Payments
- Reports
- Users and settings

13. Understand the Staff Reservation Workflow

```
New Booking
↓
Staff reviews reservation
↓
Guest arrives
↓
Check-in
↓
Guest stays
↓
Check-out
↓
Room becomes available for cleaning
↓
Cleaning completed
↓
Room becomes Available
```

Manual Booking

Staff should be able to create a reservation manually for phone, walk-in or other channels.

```
Phone / Walk-in / Other Channel
↓
Staff creates booking
↓
Availability checked
↓
Booking created
↓
Payment status recorded
```

14. Understand the Administrator Workflow

```
Administrator
■
```

- Manage Room Types
- ■■■ Manage Physical Rooms
-
- Manage Rates
- Manage Promotions
- Manage Reservations
- Manage Users
- Manage Website Content
- View Reports

15. Think About Edge Cases

A strong developer does not only think about the normal flow. The intern should identify what can go wrong.

Scenario	Expected Question
Check-out before check-in	Should the system reject the search?
No rooms available	What should the guest see?
Room becomes unavailable during checkout	Does the backend re-check availability?
Payment fails	Should the booking be confirmed?
Guest refreshes payment page	Can payment be processed twice?
Staff cancels a booking	What happens to payment/refund?
Room is under maintenance	Should it appear as available?
Promo code expired	Should discount be applied?
Two users book the last room	How is double booking prevented?

16. Phase 1 Practical Exercises

Exercise A — Draw the Guest Journey

Create a flowchart showing every step from visiting the website to receiving a confirmed booking.

Exercise B — Draw the Staff Journey

Create a flowchart showing a reservation from creation to check-out.

Exercise C — Identify Entities

List all business objects that will need to exist in the database. Do not design tables yet; focus on business understanding.

Exercise D — Identify Business Rules

Write at least 15 rules the application must enforce.

Exercise E — Identify Edge Cases

Write at least 10 things that could go wrong in the booking process and what the system should do.

Exercise F — Explain the Difference

- Room Type vs Room
- Room Status vs Booking Status

- Booking Status vs Payment Status
- Guest vs User
- Availability vs Reservation

17. Phase 1 Deliverables

Deliverable	Description
1. Product Summary	One-page explanation of what the Santo Hotel system does.
2. User Map	Guest, Staff and Administrator responsibilities.
3. Guest Flow	Complete booking journey diagram.
4. Staff Flow	Reservation/check-in/check-out workflow diagram.
5. Entity List	Business objects identified before database design.
6. Business Rules	At least 15 important rules.
7. Edge Cases	At least 10 potential failure/exception scenarios.
8. Public/Private Map	Which screens/actions require authentication.
9. Questions & Assumptions	Unclear requirements needing confirmation.

18. Mentor Review Questions

1. What problem is Santo Hotel solving with this application?
2. What is the difference between a room type and a physical room?
3. How does the system determine whether a room is available?
4. What happens if two guests try to book the last room?
5. Why should the backend calculate the final booking price?
6. What is the difference between booking status and payment status?
7. Which users can access the administration dashboard?
8. Why does the system need manual booking functionality?
9. What happens if payment fails?
10. What happens when a room is placed under maintenance?
11. Which business rules should never be trusted from the frontend?
12. What information does a booking need?
13. What questions about the hotel business are still unanswered?

19. Phase Completion Gate

The intern should not move to Phase 2 simply because the exercises were submitted. The mentor should verify understanding.

- Intern can explain the product without reading the PRD.
- Intern can draw the guest booking workflow correctly.
- Intern understands room type versus physical room.

- Intern understands the three major status groups.
- Intern can identify major backend business rules.
- Intern can identify at least 10 realistic edge cases.
- Intern has documented unresolved business questions.
- Intern can explain why availability must be checked on the backend.

PHASE 1 COMPLETE — Once the intern can explain the system clearly and the mentor approves the deliverables, proceed to Phase 2: Database & Domain Model.

20. Questions to Confirm Before Coding

Some hotel policies cannot safely be invented by the developer. Confirm these with Santo Hotel before turning them into code.

- What currency will Santo Hotel use?
- What are the official check-in and check-out times?
- What is the cancellation policy?
- What is the no-show policy?
- Are taxes included in displayed prices?
- Are there additional service fees?
- Which payment gateway will be used?
- Can guests book without an account?
- How are children counted for occupancy?
- Can one booking contain multiple rooms?
- What identification information is required?
- What happens if payment succeeds but booking creation fails?
- Who can cancel or modify reservations?
- Who can change room prices?
- What notification channels are required?
- What rooms or dates have special pricing?

21. The Mental Model

```

BUSINESS
↓
Requirements
↓
Users & Workflows
↓
Business Rules
↓
Entities / Data
↓
Backend Services
↓
API
↓
Frontend
↓
Testing

```



Deployment

The key lesson is that software development starts with understanding the business problem. The code comes after the workflow is understood.

FINAL PRINCIPLE — If the intern cannot explain the business process, they are not ready to implement the business logic.